

Statistics I

CSIT — 2nd Semester — 2082 — Answer Sheet
Subject Code: STA164

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. What do you mean by statistics? Describe briefly. The following are the marks of 30 students in Statistics. 60 54 22 36 49 84 79 56 28 43 80 50 60 56 57 46 39 57 73 65 69 48 75 65 59 73 46 44 70 22. By taking suitable class interval find Karl Pearson's coefficient of skewness interpret the results.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What do you understand by correlation analysis? Bradford Electric Illuminating is studying the relationship between kilowatt ■ hours (thousands) used and the number of rooms in a private single-family residence. A random sample of 5 houses yields the following.

Number of rooms	Kilowatts ■ hours (thousands) used
11	97
13	124
17	148
10	97
10	124

(i) Find the correlation coefficient between number of rooms and kilowatts ■ hours used. Also Interpret the result. (ii) Determine the regression equation of kilowatts hours used on number of rooms. (iii) Estimate the kilowatts -hours used for a 8 rooms house.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Two continuous random variables X and Y have following pdf: $f(x, y) = k \times y$, $0 \leq y \leq 1$, $0 \leq x \leq 1$, otherwise 0. Determine (i) the value of k (ii) marginal pdf of X and Y (iii) whether X and Y are independent?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Maximal static in respiratory pressure is an index of respiratory muscle strength. The following data show the measure of maximal static in respiratory for 11 cystic fibrosis patients: 115, 95, 100, 85, 90, 70, 45, 115, 40, 115, and 95. (i) Calculate five number summary. (ii) Construct a box- and ■whisker plot. Also comment on the shape of the distribution.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Calculate an appropriate measure of central tendency from the following table. Give the reason for your choice of measure of central tendency.

Age in years	Below 20	20-30	30-40	40-50	50 and more
Number of persons	122	72	51	51	11

Also find the lowest age of oldest 20% persons.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define independent events in probability. A problem of statistics is given to three students A, B and C whose chances of solving the problem are in the ratio 2: 3: 5. Find the probability that (i) all of them solve the problem. (ii) the problem will be solved.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. State Bayes' theorem of probability. A computer maker receives parts from three suppliers S1, S2, and S3. Fifty percent come from S1, thirty percent from S2, and twenty percent from S3. Among all the parts supplied by S1, 5% are defective, those supplied by S2, 3% are defective and 4% respectively. A customer complains that a certain part in his recently purchased computer is defective. What is the probability that it was supplied by S1?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is sampling? Give main objects of sampling.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Under what conditions the binomial distribution is used? A multiple choice test has 5 questions. There are 4 choices for each question of which one is correct. A student who has not studied for the test decides to answer all the questions randomly. What is the probability that he will get (i) all questions correct (ii) at least four questions correct?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Under what conditions binomial distribution tends to poisson distribution? If the probability that an individual suffers an adverse reaction from a particular drug is known to be 0.0001, determine the probability that out of 20000 individuals (i) exactly 3 individual will suffer an adverse reaction (ii) more than one individual will suffer an adverse reaction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. The finished inside diameter of a piston ring is normally distributed with a mean of 10 cm and standard deviation of 0.04 cm. (i) What proportion of rings will have an inside diameter exceeding 10.08 cm (ii) Below what value of inside diameter will 10% of the piston rings fall?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Given the following bivariate probability distribution:

X\Y	0	1	2
0	10	15	7
1	15	13	12
2	15	22	15

Obtain (i) marginal probability distribution of X (ii) $P(X < 0, Y = 2)$ (iii) conditional probability distribution of X given $Y = 0$. OR Write short notes on any two. (i) Primary data and secondary data (ii) Kurtosis (iii) Regression analysis.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.